

# OmniVR audiovisual data pipeline

☐ Clean / target    ☒ Processing    ☐ LQ

## (a) Collect & screen



# 100K

3–20 s

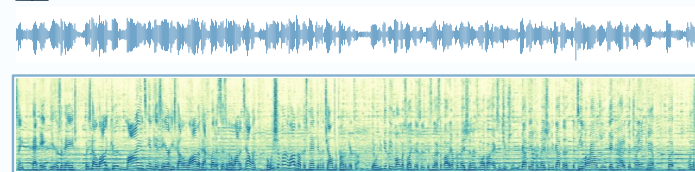
high-quality Internet clips

Example: 1080p · 30 fps · 19.93 s



### Clean audio

44.1 kHz · stereo



0–19.9 s · 0–8 kHz



### Quality & safety

**Text & overlays**

OCR · watermark · border · UGC

**Visual quality**

DOVER · aesthetic · sharpness

**Safety & motion**

NSFW · flow / SSIM / VMAF

**Color & artifacts**

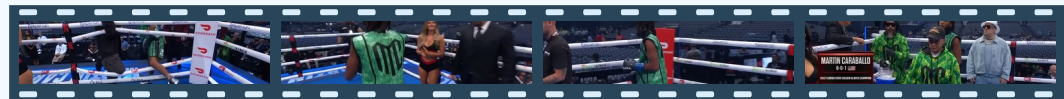
Saturation · mosaic checks

## (b) Align, caption & degrade



### Shot cutting

TransNetV2 / PySceneDetect



Same shot · temporal frame samples

Audio retained · early anchor

Cut context + selected frames + ASR



### Gemini · full-shot captioning

Frames + audio + cut context + extracted evidence



*caption\_en · excerpt*

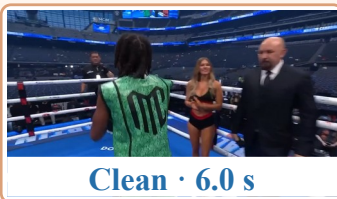
```
{
  "Summary": "A boxer circles the ring pre-fight",
  "Roles_and_Subjects":
    "Boxer ID_A; referee ID_B; ring girls",
  "Style": "Blue / neon green sports broadcast",
  "Camera_Movement":
    "Handheld tracking with close-ups",
  "Background": "Sponsor ropes + stadium",

```

```
"BGM": "Electronic music with heavy bass",
"Sound_Effects": "Ambient crowd murmur",
"Action_and_Dialogue":
  "Adjusts gloves, then walks clockwise",
"Dialogue_Timestamps": [{
  "time": "0:02–0:05", "role": "ID_B",
  "content": "Stay sharp, keep your feet moving!"
}],
"Audio_Language": "English"
}
```



### Visual degradation



Clean · 6.0 s



LQ + grayscale

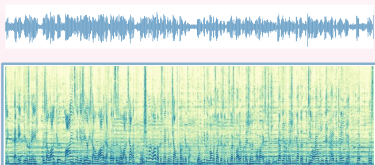
6× downsample · x265 CRF 31 · grayscale

Vertical scratches · dust · soft bands



### Acoustic degradation

Clean



Clean · 6.0 s

Degraded



Filter · noise · clipping · MP3 · reverb

Noise SNR 13.0 dB · clip threshold 0.351

## (c) Paired supervision



### Degraded conditions

grayscale



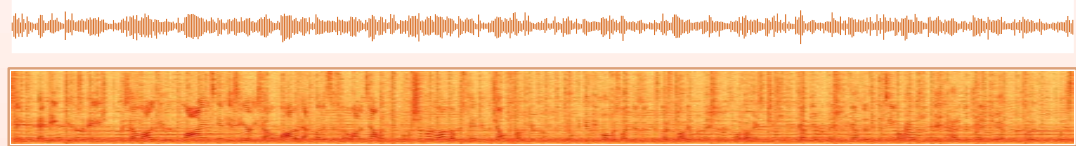
00:00.7



00:06.0

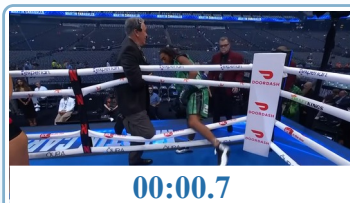


00:12.0

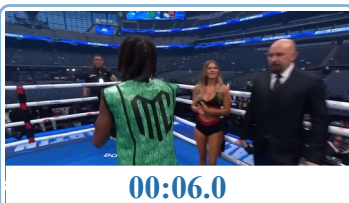


### Clean targets

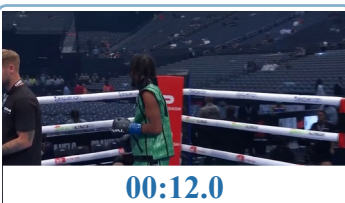
color + detail



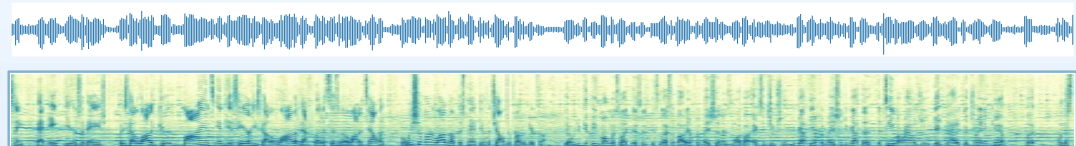
00:00.7



00:06.0



00:12.0



### Aligned training record

Shot ID · timestamps · caption JSON · operator metadata



### HR audiovisual refiner

LQ conditions + clean targets · second-stage training